COMPUTER ENGINEERING

QUEZON CITY

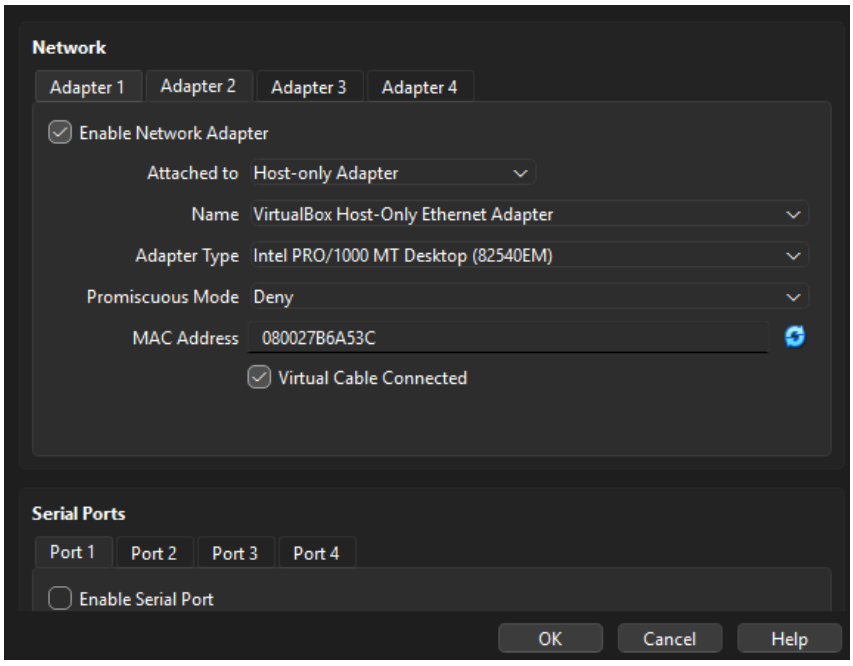
COURSE CODE/TITLE:	CPE 212 Automating Server Management	**SECTION:**	CPE31s2
DATE PERFORMED:	09/07/2026	**DATE SUBMITTED:**	20/07/2026
NAME:	Damie Jake D. Gestiada	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. 1.1 Creating Virtual Machines

1. PROCEDURE OUTPUT

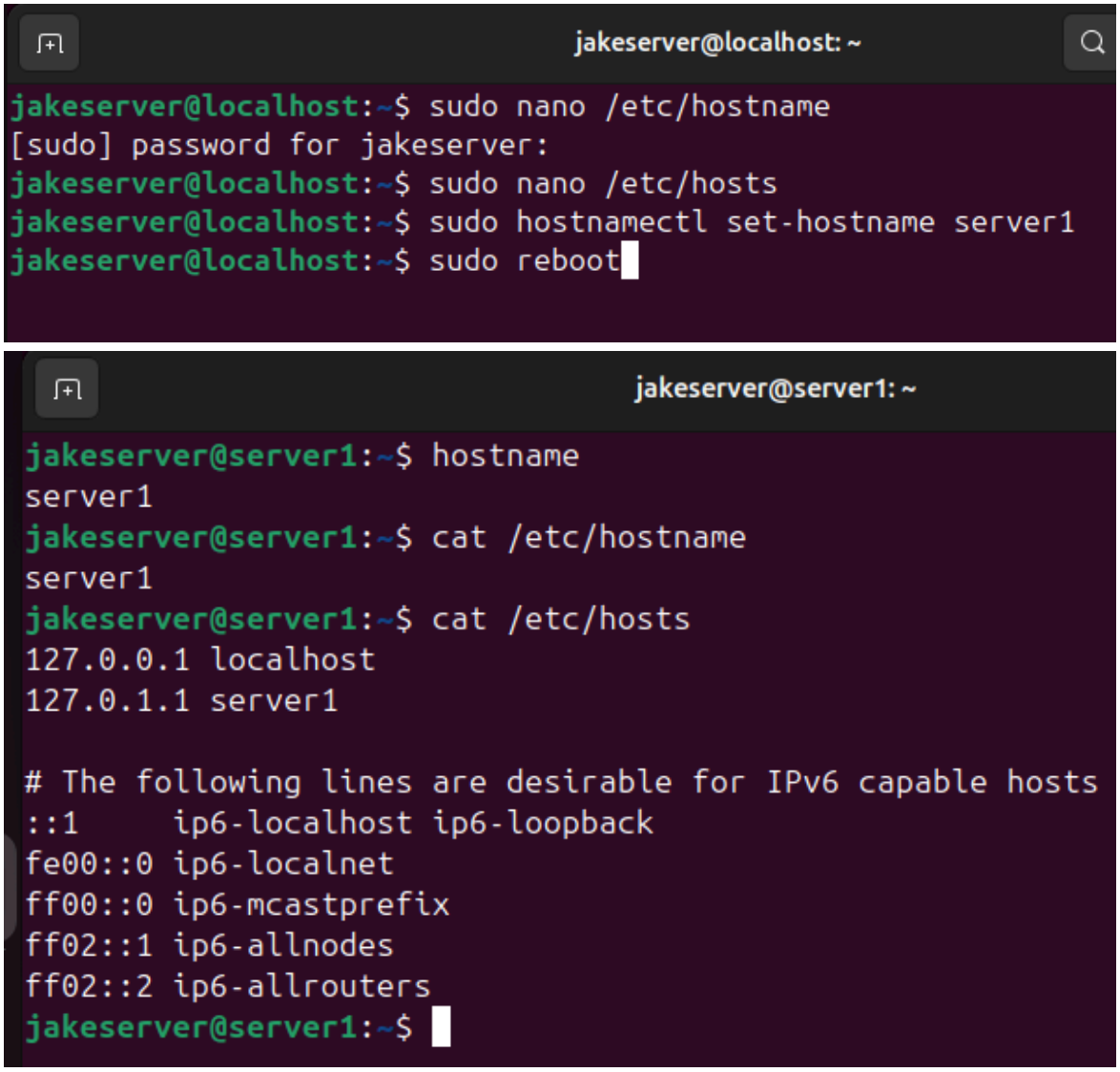
What procedures did you perform?

Task 1:
Do the following on Server 1, Server 2, and Local Machine. In editing the file using nano command, press control + O to write out (save the file). Press enter when asked for the name of the file. Press control + X to end.

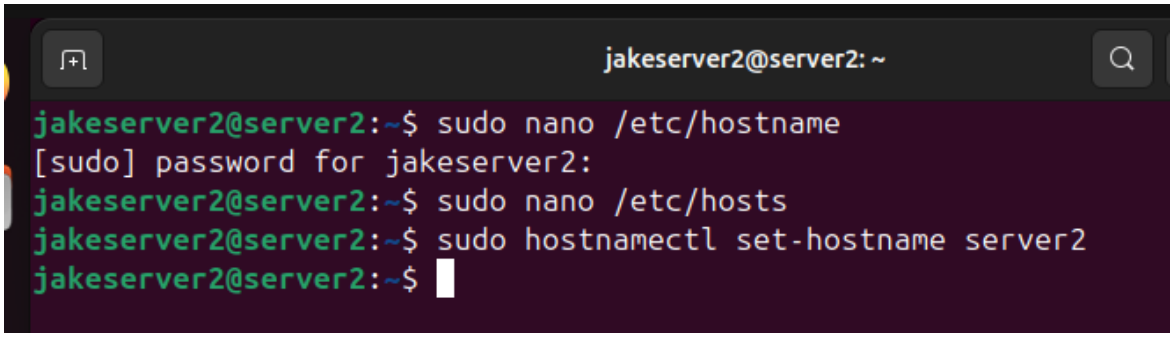


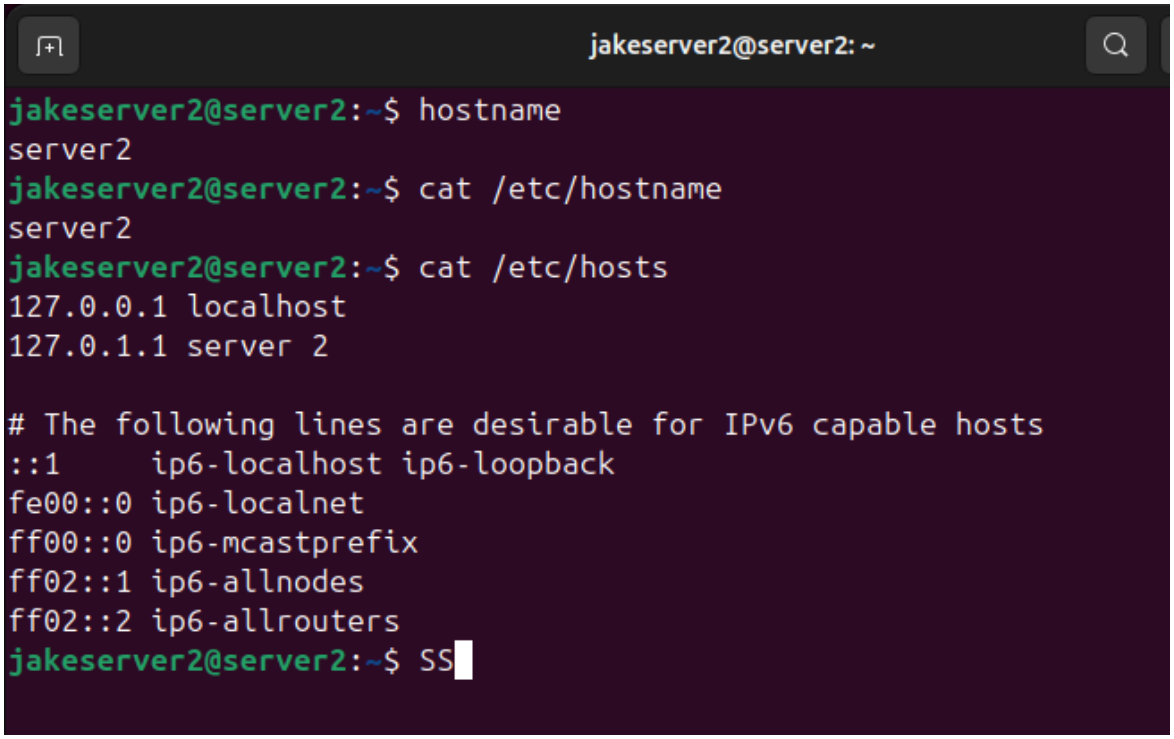
1. Change the hostname using the command sudo nano /etc/hostname
1.1 Use server1 for Server 1
1.2 Use server2 for Server 2
1.3 Use workstation for the Local Machine

1.
Server1

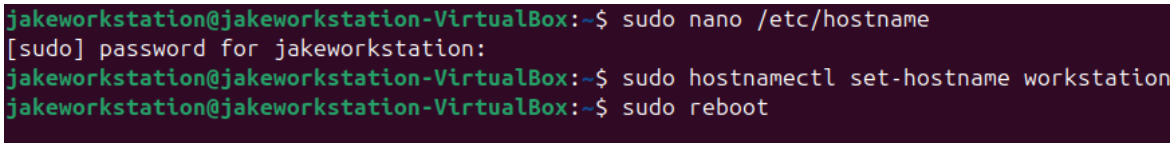


Server2





Workstation



```
jakeworkstation@workstation:~$ hostname
workstation
jakeworkstation@workstation:~$ cat /etc/hostname
workstation
jakeworkstation@workstation:~$ cat /etc/hosts
127.0.0.1 localhost
127.0.1.1 workstation

# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
jakeworkstation@workstation:~$
```

2. Edit the hosts using the command sudo nano /etc/hosts. Edit the second line.
2.1 Type 127.0.0.1 server 1 for Server 1
2.2 Type 127.0.0.1 server 2 for Server 2
2.3 Type 127.0.0.1 workstation for the Local Machine

Task 2:
Configure SSH on Server 1, Server 2, and Local Machine. Do the following:
1. Upgrade the packages by issuing the command sudo apt update and sudo apt upgrade respectively.

2. Install the SSH server using the command sudo apt install openssh-server.
3. Verify if the SSH service has started by issuing the following commands:
3.1 sudo service ssh start
3.2 sudo systemctl status ssh

4. Configure the firewall to all port 22 by issuing the following commands:
4.1 sudo ufw allow ssh
4.2 sudo ufw enable
4.3 sudo ufw status

```
jakeserver@server1:~$ sudo service ssh start
jakeserver@server1:~$ sudo systemctl start ssh
jakeserver@server1:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/system
d/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
Created symlink /etc/systemd/system/sshd.service → /usr/lib/systemd/system/ssh.s
ervice.
Created symlink /etc/systemd/system/multi-user.target.wants/ssh.service → /usr/l
ib/systemd/system/ssh.service.
```

```
jakeserver@server1:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enab>
     Active: active (running) since Tue 2026-07-14 00:56:26 PST; 4min 1s ago
TriggeredBy: ● ssh.socket
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 19989 (sshd)
      Tasks: 1 (limit: 6119)
     Memory: 1.2M (peak: 1.5M)
        CPU: 14ms
     CGroup: /system.slice/ssh.service
             └─19989 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"
Trash

Jul 14 00:56:25 server1 systemd[1]: Starting ssh.service - OpenBSD Secure Shell>
Jul 14 00:56:26 server1 sshd[19989]: Server listening on 0.0.0.0 port 22.
Jul 14 00:56:26 server1 sshd[19989]: Server listening on :: port 22.
Jul 14 00:56:26 server1 systemd[1]: Started ssh.service - OpenBSD Secure Shell >
lines 1-17/17 (END)
```

4.

```
jakeserver@server1:~$ sudo ufw allow ssh
Rules updated
Rules updated (v6)
jakeserver@server1:~$ sudo ufw enable
Firewall is active and enabled on system startup
jakeserver@server1:~$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW       Anywhere
22/tcp (v6)                ALLOW       Anywhere (v6)

jakeserver@server1:~$ sudo ss -tulpn | grep ':22'
tcp   LISTEN 0      4096         0.0.0.0:22        0.0.0.0:*    users:(("sshd",
pid=19989,fd=3),("systemd",pid=1,fd=344))
tcp   LISTEN 0      4096            [::]:22           [::]:*    users:(("sshd",
pid=19989,fd=4),("systemd",pid=1,fd=345))
```

Task 3:
Verify network settings on Server 1, Server 2, and Local Machine. On each device, do the following:
1. Record the ip address of Server 1, Server 2, and Local Machine. Issue the command ifconfig and check network settings. Note that the ip addresses of all The machines are in this network 192.168.56.XX.
1.1 Server 1 IP address: 192.168.56.101
1.2 Server 2 IP address: 192.168.56.102
1.3 Server 3 IP address: 192.168.56.103
2. Make sure that they can ping each other.

2.1 Connectivity test for Local Machine 1 to Server 1: ☐✔ Successful ☐ Not Successful
2.2 Connectivity test for Local Machine 1 to Server 2: ☐✔ Successful ☐ Not Successful
2.3 Connectivity test for Server 1 to Server 2: ✔ Successful ☐ Not Successful

Server1:

```
    link/ether 08:00:27:97:cf:6a brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.102/24 brd 192.168.56.255 scope global dynamic noprefixrout
 enp0s8
       valid_lft 587sec preferred_lft 587sec
    inet6 fe80::b7a8:a189:5700:f241/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
jakeserver@server1:~$ hostname -I
10.0.2.15 192.168.56.102 fd17:625c:f037:2:d30:2bab:913f:7e66 fd17:625c:f037:2:a
0:27ff:fe43:3215
jakeserver@server1:~$
```

Server2 IPv4:

```
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gr
oup default qlen 1000
    link/ether 08:00:27:b6:a5:3c brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.103/24 brd 192.168.56.255 scope global dynamic noprefixroute
 enp0s8
       valid_lft 474sec preferred_lft 474sec
    inet6 fe80::6b27:f7d3:332f:f47e/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
jakeserver2@server2:~$ hostname -I
10.0.2.15 192.168.56.103 fd17:625c:f037:2:d62d:2207:157f:6873 fd17:625c:f037:2:a
00:27ff:fe34:48b6
jakeserver2@server2:~$
```

Workstation IPv4:

```
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gr
oup default qlen 1000
    link/ether 08:00:27:e6:73:a8 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.101/24 brd 192.168.56.255 scope global dynamic noprefixroute
 enp0s8
       valid_lft 502sec preferred_lft 502sec
    inet6 fe80::3b4d:4036:bcf8:b898/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
jakeworkstation@workstation:~$ hostname -I
10.0.2.15 192.168.56.101 fd17:625c:f037:2:eaf8:a27b:1ee:ff2b fd17:625c:f037:2:a0
0:27ff:fe33:bb14
jakeworkstation@workstation:~$
```

Ping

```
jakeworkstation@workstation:~$ ping -c 4 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=0.467 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=0.234 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=0.224 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=0.249 ms

--- 192.168.56.102 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3230ms
rtt min/avg/max/mdev = 0.224/0.293/0.467/0.100 ms
```

```
jakeworkstation@workstation:~$ ping -c 4 192.168.56.103
PING 192.168.56.103 (192.168.56.103) 56(84) bytes of data.
64 bytes from 192.168.56.103: icmp_seq=1 ttl=64 time=0.382 ms
64 bytes from 192.168.56.103: icmp_seq=2 ttl=64 time=0.236 ms
64 bytes from 192.168.56.103: icmp_seq=3 ttl=64 time=0.229 ms
64 bytes from 192.168.56.103: icmp_seq=4 ttl=64 time=0.228 ms

--- 192.168.56.103 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3057ms
rtt min/avg/max/mdev = 0.228/0.268/0.382/0.065 ms
jakeworkstation@workstation:~$
```

```
jakeserver@server1:~$ ping -c 4 192.168.56.103
PING 192.168.56.103 (192.168.56.103) 56(84) bytes of data.
64 bytes from 192.168.56.103: icmp_seq=1 ttl=64 time=0.401 ms
64 bytes from 192.168.56.103: icmp_seq=2 ttl=64 time=0.240 ms
64 bytes from 192.168.56.103: icmp_seq=3 ttl=64 time=0.214 ms
64 bytes from 192.168.56.103: icmp_seq=4 ttl=64 time=0.233 ms

--- 192.168.56.103 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3316ms
rtt min/avg/max/mdev = 0.214/0.272/0.401/0.075 ms
jakeserver@server1:~$
```

Task 4:
Verify SSH connectivity on Server 1, Server 2, and Local Machine.
1. On the Local Machine, issue the following commands:
1.1 ssh username@ip_address_server1 for example, ssh jvtaylar@192.168.56.120
1.2 Enter the password for server 1 when prompted
1.3 Verify that you are on server 1. The user should be in this format user@server1.
For example, jvtaylar@server1

2. Logout of Server 1 by issuing the command control + D.
3. Do the same for Server 2.
4. Edit the hosts of the Local Machine by issuing the command sudo nano /etc/hosts. Below all texts type the following:
4.1 IP_address server 1 (provide the ip address of server 1 followed by the hostname)
4.2 IP_address server 2 (provide the ip address of server 2 followed by the hostname)
4.3 Save the file and exit.

5. On the local machine, verify that you can do the SSH command but this time, use the hostname instead of typing the IP address of the servers. For example, try to do ssh jvtaylar@server1. Enter the password when prompted. Verify that you have entered Server 1. Do the same for Server 2

Server1

```
jakeworkstation@workstation:~$ whoami
jakeworkstation
jakeworkstation@workstation:~$ ssh jakeserver@192.168.56.102
jakeserver@server1:~$ hostname
server1
jakeserver@server1:~$ whoami
jakeserver
jakeserver@server1:~$ hostname -I
10.0.2.15 192.168.56.102 fd17:625c:f037:2:d30:2bab:913f:7e66 fd17:625c:f037:2:a0
0:27ff:fe43:3215
jakeserver@server1:~$
```

Server2

```
jakeworkstation@workstation:~$ ssh jakeserver2@192.168.56.103
jakeserver2@192.168.56.103's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-35-generic x86_64)
jakeserver2@server2:~$ whoami
jakeserver2
jakeserver2@server2:~$ hostname
server2
jakeserver2@server2:~$ hostname -I
10.0.2.15 192.168.56.103 fd17:625c:f037:2:37fe:d96:a70f:28a7 fd17:625c:f037:2:a0
0:27ff:fe34:48b6
jakeserver2@server2:~$
```

```
jakeworkstation@workstation: ~
  GNU nano 7.2                                /etc/hosts
127.0.0.1 localhost
127.0.1.1 workstation

198.168.56.102 server1
198.168.56.103 server2

# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
```

```
jakeworkstation@workstation:~$ getent hosts server1
198.168.56.102  server1
jakeworkstation@workstation:~$ getent hosts server2
198.168.56.103  server2
jakeworkstation@workstation:~$ ping -c 4 server1
PING server1 (198.168.56.102) 56(84) bytes of data.


--- server1 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 8833ms

jakeworkstation@workstation:~$
jakeworkstation@workstation:~$ ping -c 4 server2
PING server2 (198.168.56.103) 56(84) bytes of data.

--- server2 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 13804ms
```

COMPUTER ENGINEERING

QUEZON CITY

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

Reflection:

Answer the following:

1. How are we able to use the hostname instead of IP address in SSH commands?
   - Our reason to use hostname is that we connected every server name with its IP address in workstation's /etc/hosts file so Ubuntu can understand server1 and server2 refers to different machines. Therefore using IP address every time while connecting via SSH becomes unnecessary for us.
2. How secured is SSH?
   - SSH is known for its security which comes from the fact that it encrypts the entire computer-to-computer conversation. The result is that username, password, commands, and data being moved are all safe from the view of other people on the network. When you use a good password or set up SSH keys and always keep your system up-to-date, SSH is going to be much more secure for you.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Gemini and Grammarly**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **For grammar correction and explanation of concept.**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.